

Foundation (Jalapeno)

Q1. Solve for x : $2x + 5 = 11$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Isolate y in $3y - 2 = 7$

- (A) $y = 3$
- (B) $y = 5$
- (C) $y = 7$
- (D) $y = 9$
- (E) $y = 11$

Q3. In basket ball, a player scores $2x + 10$ points in the first half and $x - 5$ points in the second half. If the total score is 35, what is x ?

- (A) $x = 5$
- (B) $x = 7$
- (C) $x = 10$
- (D) $x = 12$
- (E) $x = 15$

Q4. A cyclist travels $2x$ miles in the morning and $x + 1$ miles in the afternoon. If the total distance is 21 miles, what is x ?

- (A) $x = 6$
- (B) $x = 7$
- (C) $x = 8$
- (D) $x = 9$
- (E) $x = 10$

Q5. Solve for z : $z/4 + 2 = 9$

- (A) $z = 20$
- (B) $z = 24$
- (C) $z = 28$
- (D) $z = 32$
- (E) $z = 36$

Q6. In a tennis match, a player wins $3x - 2$ games in the first set and $2x + 1$ games in the second set. If the total number of games won is 16, what is x ?

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q7. Isolate w in $w/2 - 3 = 7$

- (A) $w = 10$
- (B) $w = 12$
- (C) $w = 14$
- (D) $w = 16$
- (E) $w = 20$

Q8. A gymnast scores $x + 2$ points for the vault and $3x - 1$ points for the floor exercise. If the total score is 20, what is x ?

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = 6$
- (D) $x = 7$
- (E) $x = 8$

Open Ended Questions (FRQ)

Q9. Solve for y in the equation $2y + 5 = 3y - 2$. Show all steps.

Q10. Isolate x in the equation $x/3 + 2 = 2x - 5$. Show all steps.

Chef's Tips

- Read each question carefully before answering.
- Use the space provided for workings.
- Check your answers.